



Xiaotao Shen, Ph.D.

Nanyang Assistant Professor

Lee Kong Chian School of Medicine, Nanyang Technological University, Singapore

59 Nanyang Dr, Singapore, 636921.

✉ xiaotao.shen@ntu.edu.sg

🏠 shen-lab.org

📞 [+65 83042333](tel:+6583042333)

🌐 [Xiaotao Shen](#)

🔗 [jaspershen](#)

Research Interests

- Multi-omics Data Integration Methods Development
- Host-Microbe Interaction
- Aging and Aging-Related Diseases
- Maternal and Child Health
- Wearable Data and Precision Medicine
- Environmental Health

Education & Research Experiences

- **Director** 2025/3-present, [Singapore Phenome Center](#)
- **Adjunct Principal Investigator** 2024/10-present, **KK Women's and Children's Hospital, Singapore**
- **Nanyang Assistant Professor** 2024/5-present, **Nanyang Technological University, Singapore**
- **Postdoctoral Research Fellow** 2019/4–2024/4, **Stanford University**
(Advisor: [Prof. Michael Snyder](#))
- **Ph.D. in Bioinformatics and Metabolomics** 2013/9–2018/12, **Chinese Academy of Sciences**
(Advisor: [Prof. Zheng-Jiang Zhu](#))
- **B.S. in Biotechnology** 2009/9–2013/6, **Inner Mongolia University**

Honors & Awards

- Young Investigator Award **The 3rd Chinese American Society For Mass Spectrometry Conference** (2023)
- SPORR 2023 Rigor and Reproducibility Award **Stanford Program on Research Rigor and Reproducibility** (2023)
- Winner Selected by the Event Committee **Exposome Data Challenge Event** (2021)
- Student Travel Award for Oral Presentation **The International Metabolomics Society** (2018)
- International Conference Travel Award **The Metabolites Journal** (2018)
- China National Scholarship **Ministry of Education of the People's Republic of China** (2017)
- Award for Outstanding Youth Report **The 3rd China MS Analysis Conference** (2017)
- Merit Student **University of Chinese Academy of Sciences** (2016)
- Award for Outstanding Youth Report **The 34th China MS Society Conference** (2016)
- Inner Mongolia Outstanding Graduate **Inner Mongolia Autonomous Region** (2013)
- National Encouragement Scholarship **Inner Mongolia University** (2011)

Corresponding/Co-corresponding Author Publications

* Co-Corresponding Author

1. Y. Liu, Y. Wang, T. Huan, **X. Shen***, FeatureMSEA: Metabolic Feature-based Metabolite Set Enrichment Analysis. **bioRxiv**, 2026.
2. Y. Ge⁺, S. Bai⁺, Z. Qiang⁺, Y. Liu⁺, Y. Wu, **X. Shen***, LagCI Enables Inference of Temporal Causal Relationships from Dense Multi-Omic Time Series. **bioRxiv**, 2026.
3. J. Zhang, C. Jiang, X. Zhou, P. Gao, S. Wong, M. Snyder, **X. Shen***, Cross-body site microbial interactions influence the human plasma metabolome. **Microbiome**, 2026. <https://doi.org/10.1186/s40168-026-02405-w>
4. X Wang⁺, Y. Liu⁺, C. Jiang, Z. Huang, H. Yan, S. Wong, C. H. Johnson, J. Zhang, Y. Ge, F. Zhang, R. Lai, P. Gao^{*}, X. Zhang^{*}, **X. Shen***, TidyMass2: Advancing LC-MS Untargeted Metabolomics Through Metabolite Origin Inference and Metabolic Feature-based Functional Module Analysis. **Nature Communications**, 2026. <https://doi.org/10.1038/s41467-026-68464-7>
5. Y. Ge⁺, F. Zhang⁺, Y. Liu⁺, C. Jiang, P. Gao, N. Tan, S., Y. Shen, Q. Zhou, X. Zhou, C. Wang^{*}, **X. Shen***, **Leveraging Large Language Models for Redundancy-Aware Pathway Analysis and Deep Biological Interpretation**. **bioRxiv**, 2025.
6. S. Liu⁺, H. Sun⁺, Z. Du⁺, S. Lu, C. Wang, Y. Zhang, Z. Luo, L. Wang, Z. Fan, P. Wei, Y. Yan, J. Zhang, S. Yin, T. Liu, Q. He, X. Guo, K. Ding, J. Zhou, H. Hua, C. Yu, W. Xu, J. Shan, Y. Li, Y. Xu, **X. Shen***, G. Cao^{*}, W. Zhou^{*}, **Metabolomics and proteomics reveal blocking argininosuccinate synthetase 1 alleviates colitis in mice**. **Nature Communications**, 2025.
7. Z. Du⁺, F. Zhang⁺, Y., Y. Liu, H. Yu, Y. Wang, R. Dalan, **X. Shen***, **Application of Wearable Devices in Diabetes Management**. **Health and Metabolism**, 2025, 7-7. [↗](#)
8. Y Liu, F Zhang, Y Ge, Q Liu, S He, **X Shen***, **Application of LLMs/Transformer-Based Models for Metabolite Annotation in Metabolomics**. **Health and Metabolism**, 2025, 7-7.

First/Co-first Author Publications

+ Co-first Author.

1. **X. Shen***, S. Chen⁺, L. Liang⁺, M. Avina, H. Zackriah, L. Jelliffe-Pawlowski, L. Rand, M. P. Snyder^{*}, **Longitudinal Urine Metabolic Profiling and Gestational Age Prediction in Pregnancy**. **Briefing in Bioinformatics**, 2025, 26 (1). [↗](#)
2. X. Zhou⁺, **X. Shen***, J. Johnson, D. Spakowicz, M. Agnello, W. Zhou, M. Avina, A. Honkala, F. Chleilat, S. Chen, K. Cha, S. Leopold, C. Zhu, L. Chen, L. Lyu, D. Hornburg, S. Wu, X. Zhang, C. Jiang, L. Jiang, L. Jiang, R. Jian, A. Brooks, M. Wang, K. Contrepois, P. Gao, S. Rose, T. Tran, H. Nguyen, A. Celli, B. Hong, E. Bautista, Y. Dorsett, P. Kavathas, Y. Zhou, E. Sodergren, G.M. Weinstock, M.P. Snyder^{*}. **Longitudinal Profiling of the Microbiome at Four Body Sites Reveals Core Stability and Individualized Dynamics During Health and Disease**. **Cell Host&Microbe**, 2024 [↗](#)
3. **X. Shen***, R. Kellogg⁺, D. Panyard⁺, N. Bararpour⁺, K. Castillo, B. Lee-McMullen, A. Delfarah, J. Ubellecker, S. Ahadi, Y. Rosenberg-Hasson, A. Ganz, K. Contrepois, B. Michael, I. Simms, C. Wang, D. Hornburg, M.P. Snyder^{*}, **Multi-omic Microsampling Captures Health Perturbations in A Lifestyle Context**. **Nature Biomedical Engineering**, 2023. [↗](#)
4. **X. Shen***, C. Wang⁺, M.P. Snyder^{*}, **massDatabase: Utilities for the Operation of the Public Compound and Pathway Database**, **Bioinformatics**, 2022, btac546. [↗](#)

5. **X. Shen**⁺, W. Shao⁺, C. Wang⁺, L. Liang, S. Chen, S. Zhang, M. Rusu⁺, M.P. Snyder⁺, **Deep Learning-based Pseudo-Mass Spectrometry Imaging Analysis for Precision Medicine**, *Briefing in Bioinformatics*, 2022, bbac331. [↗](#)
6. **X. Shen**⁺, H. Yan⁺, C. Wang⁺, P. Gao, C.H. Johnson⁺, M.P. Snyder⁺, **TidyMass an Object-oriented Reproducible Analysis Framework for LC-MS Data**, *Nature Communications*, 2022, 4365. [↗](#)
7. P. Gao⁺, **X. Shen**⁺, X. Zhang, C. Jiang, M. P. Snyder⁺, **Precision Environmental Health Monitoring by Longitudinal Exposome and Multi-omics Profiling**, *Genome Research*, 2022, **32**, 1199-1214. [↗](#)
8. **X. Shen**⁺, S. Wu⁺, L. Liang, S. Chen, K. Contrepois, Z.-J. Zhu⁺ and M.J. Snyder⁺, **metID: an R Package for Automatable Compound Annotation for LC-MS-based Data**, *Bioinformatics*, 2021, **1**, 1-2. [↗](#)
9. **X. Shen**, R. Wang, X. Xiong, Y. Yin, Y. Cai, J. Ma, N. Liu, Z.-J. Zhu⁺, **Large-scale Metabolite Identification for Untargeted Metabolomics Using Metabolic Reaction Network**, *Nature Communications*, 2019, 10:1516. [↗](#)
10. **X. Shen** and Z.-J. Zhu⁺, **MetFlow: An Interactive and Integrated Workflow For Metabolomics Data Cleaning and Differential Metabolite Discovery**, *Bioinformatics*, 2019, **35**, 16. [↗](#)
11. H. Jia⁺, **X. Shen**⁺, Y. Guan, M. Xu, M. Mo, J. Zhu⁺ and Z.-J. Zhu⁺, **Predicting the Pathological Response to Neoadjuvant Chemoradiation Using Untargeted Metabolomics in Locally Advanced Rectal Cancer**, *Radiotherapy and Oncology*, 2018, **128**, 548-556. [↗](#)
12. J. Wang⁺, T. Zhang⁺, **X. Shen**⁺, J. Liu, D. Zhao, Y. Sun, L. Wang, Y. Liu, X. Gong, Y. Liu, Z.-J. Zhu⁺, F. Xue⁺, **Serum Metabolomics for Early Diagnosis of Esophageal Squamous Cell Carcinoma by UHPLC-QTOF/MS**, *Metabolomics*, 2016, **12**: 116. [↗](#)
13. **X. Shen**, X. Gong, Y. Cai, Y. Guo, J. Tu, T. Zhang, J. Wang, F. Xue, Z.-J. Zhu⁺, **Normalization and Integration of Large-Scale Metabolomics Data Using Support Vector Regression**, *Metabolomics*, 2016, **12**: 89. [↗](#)

Collaborative Publications

1. S. Chen, Y. Yuan, Y. Wang, Y. Peng, H. Tun, Z. Jiang, Y. Miao, S. Lee, X. Yin, **X. Shen**, O. DeLeon, E. Chang, F. Chan, Y. Sun, S. Ng, Q. Su. Identification of antimicrobial peptides from ancient gut microbiomes. *Nature Communications*, 2026. <https://doi-org.stanford.idm.oclc.org/10.1038/s41467-026-68495-0>
2. Z. Song, M. Kim, J. Lee, T. Kwon, J. Kim, J. Kim, S. Kim, S. Kim, B. Jun, S. Lee, W. Park, I. Sonu, M. Rosen, C. Ong, **X. Shen**, L. Ng, G. Kim, S. Chae, K. Siah, D. Ho, S. Song, N. Martinez-Martin, J. Song, K. Yew, M. Kim, H. Huang, S. Wong, B. Lee, S. Park. Deployment of a cloud-based passive defecation monitoring system for continuous gut health monitoring. *Nature Protocols*, 2026. <https://doi-org.stanford.idm.oclc.org/10.1038/s41596-025-01296-9>
3. I. Marić, A. Mahzarnia, H. Mujuru, G. Chimhini, S. Saha, M. Hassan, N. Otieno, S. Hawken, K. Wilson, **X. Shen**, S. Lancaster, R. Wong, J. Reiss, J. Kerner, M. Snyder, W. Hay, G. Shaw, D. Stevenson, V. Ward, G. Darmstadt. *Science Advances*, 2025, **11**. 10.1126/sciadv.adu9145
4. M. Gladding, **X. Shen**, M. Snyder, P. Havel, S. Adams. Interindividual Variability in Postprandial Plasma Fructose Patterns in Adults. *Nutrients*, 2024, **16** (18), 3079.
5. E. Xie, **X. Shen**, Y. Yeo, Z. Xing, J. Ebinger, Y. Duan, Y. Zhang, S. Cheng, F. Ji, J. Deng. Exploring the underlying molecular mechanisms of acute myocardial infarction after SARS-CoV-2 infection. *American Heart Journal Plus: Cardiology Research and Practice*, 2024, **44**, 100417.
6. C. Peng, Q. Chen, S. Tan, **X. Shen**, C. Jiang. **Generalized Reporter Score-based Enrichment Analysis for Omics Data**, *Briefings in Bioinformatics*, Volume 25, Issue 3, May 2024, bbae116. [↗](#)

7. D. Ding, **X. Shen**, M.P. Snyder, R. Tibshirani, **Semi-supervised Cooperative Learning for Multiomics Data Fusion**, Machine Learning for Multimodal Healthcare Data. ML4MHD 2023. *Lecture Notes in Computer Science*, vol 14315. Springer, Cham. [↗](#)
8. S. Jain, L. Pei, J. Spraggins, M. Angelo, J. Carson, N. Gehlenborg, F. Ginty, J. Gonçalves, J. Hagood, J. Hickey, N. Kelleher, L. Laurent, S. Lin, Y. Lin, H. Liu, A. Naba, E. Nakayasu, W. Qian, A. Radtke, P. Robson, B. Stockwell, R. Plas, I. Vlachos, M. Zhou, **HuBMAP Consortium**, K. Borner, M. Snyder, **HuBMAP Consortium**, **Advances and Perspectives for the Human BioMolecular Atlas Program (HuBMAP)**, *Nature Cell Biology*, 2023. [↗](#)
9. W. Wei, N. Riley, X. Lyu, **X. Shen**, J. Guo, S. Raun, M. Zhao, M. Moya-Garzon, H. Basu, A. Tung, V. Li, W. Huang, A. Wiggenhorn, K. Svensson, M. Snyder, C. Bertozzi, J. Long, **Organism-wide, Cell-type-specific Secretome Mapping of Exercise Training in Mice**, *Cell Metabolism*, 35: 1-19, 2023. [↗](#)
10. S. Zhang, J. Cooper-Knock, A. Weimer, M. Shi, L. Kozhaya, D. Unutmaz, C. Harvey, T. Julian, S. Furini, E. Frullanti, F. Fava, A. Renieri, P. Gao, **X. Shen**, I. Timpanaro, K. Kenna, J. Baillie, M. Davis, P. Tsao, M.P. Snyder, **Multiomic Analysis Reveals Cell-type-specific Molecular Determinants of COVID-19 Severity**, *Cell Systems*, 13(8):598-614, 2022. [↗](#)
11. L. Maitre, J. Guimbaud, C. Warembourg, N. Güil-Oumrait, P. Petrone, M. Chadeau-Hyam, M. Vrijheid, X. Basagaña, J. Gonzalez, **Exposome Data Challenge Participant Consortium**, **State-of-the-art methods for exposure-health studies: Results from the exposome data challenge event**, *Environment International*, 168: 107422, 2022. [↗](#)
12. M. Wei, L. Zhao, J. Lv, X. Li, G. Zhou, B. Fan, **X. Shen**, D. Zhao, F. Xue, J. Wang, T. Zhang, **The Mediation Effect of Serum Metabolites On The Relationship Between Long-Term Smoking Exposure and Esophageal Squamous Cell Carcinoma**, *BMC cancer*, 21, 415. [↗](#)
13. J. Lv, J. Wang, **X. Shen**, J. Liu, D. Zhao, X. Li, B. Fan, Y. Sun, F. Xue, Z.-J. Zhu, T. Zhang, **A Serum Metabolomics Analysis Reveals A Panel of Screening Metabolic Biomarkers for Esophageal Squamous Cell Carcinoma**, *Clinical and Translational Medicine*, 2021, 11, 5. [↗](#)
14. X. Li, L. Zhao, M. Wei, J. Lv, Y. Sun, **X. Shen**, D. Zhao, F. Xue, T. Zhang, J. Wang, **Serum Metabolomics Analysis for The Progression of Esophageal Squamous Cell Carcinoma**, *Journal of Cancer*, 2021, 12, 3190–3197. [↗](#)
15. L. Liang, M. Rasmussen, B. Piening, **X. Shen**, S. Chen, H. Rost, J. Snyder, R. Tibshirani, L. Skotte, N. Lee, K. Contrepois, B. Feenstra, H. Zackariah, M.J. Snyder, M. Melbye, **Metabolic Dynamics and Prediction of Gestational Age and Time to Delivery in Pregnant Women**, *Cell*, 2020, 181, 7, 1680-1692. [↗](#)
16. Z. Wang, B. Cui, F. Zhang, Y. Yang, **X. Shen**, Z. Li, W. Zhao, Y. Zhang, K. Deng, Z. Rong, K. Yang, X. Yu, K. Li, P. Han, and Z.-J. Zhu, **Development of A Correlative Strategy to Discover Colorectal Tumor Tissue Derived Metabolite Biomarkers in Plasma Using Untargeted Metabolomics**, *Analytical Chemistry*, 2019, 91, 3, 2401-2408. [↗](#)
17. Z. Zhou, **X. Shen**, X. Chen, J. Tu, X. Xiong, and Z.-J. Zhu, **LipidIMMS Analyzer: Integrating Multi-dimensional Information to Support Lipid Identification in Ion Mobility–Mass Spectrometry based Lipidomics**, *Bioinformatics*, 2018, 35, 4, 698-700. [↗](#)
18. Z. Zhou, J. Tu, X. Xiong, **X. Shen**, and Z.-J. Zhu, **LipidCCS: Prediction of Collision Cross-Section Values for Lipids with High Precision to Support Ion Mobility-Mass Spectrometry based Lipidomics**, *Analytical Chemistry*, 2017, 89, 9559–9566. [↗](#)
19. Z. Zhou, **X. Shen**, J. Tu, and Z.-J. Zhu, **Large-Scale Prediction of Collision Cross-Section Values for Metabolites in Ion Mobility-Mass Spectrometry**, *Analytical Chemistry*, 2016, 88, 11084-11091. [↗](#)

Patents

1. L. Liang, M.P. Snyder, **X. Shen**, S. Chen, **Systems and Methods for Evaluating Gestational Progress and Applications Thereof**, 2023, 18251702, USA.
2. M.P. Snyder, R. Kellogg, **X. Shen**, **Multi-Omic Sample Analysis Methods**, 2022, USA.
3. Z-J. Zhu, **X. Shen**, **A Method for Metabolite Annotation and Dysregulated Pathway Analysis**, 2019, China.

Invited Talks

1. **Multi-omics and Its Application in Precision Medicine and Biomedical Studies**, 2025/7/11, The 5th Environmental Exposure and Health International Symposium, Guiyang, China.
2. **Multi-omics and its application in precision medicine and biomedical studies**, 2024/12/4, FEBS-IUBMB-ENABLE 2024 Conference, Singapore.
3. **Unraveling the Comprehensive Metabolic Network Shaped by the Gut Microbiome and Its Role in Cardiovascular Diseases**, 2024/11/25, ILSI Southeast Asia Region Symposium, Malaysia.
4. **Dynamic Changes During Human Aging Revealed in Multi-omics Profiles. 2024 Chinese Biophysics Congress**, 2024/7, Lanzhou, China.
5. **Multi-omics For Precision Medicine. Hong Kong Baptist University**, 2024/7, Hong Kong, China.
6. **Multi-omics For Precision Medicine. Metabolomics in Human Health**, 2024/5, Lee Kong Chian School of Medicine, Nanyang Technological University, Singapore.
7. **R for Mass Spectrometry Data Processing. CAS Center for Excellence in Molecular Plant Sciences**. 2024/5, Shanghai, China.
8. **Multi-omics For Precision Medicine. Interdisciplinary Research Center on Biology and Chemistry**, 2024/3, Shanghai, China.
9. **Remote and Flexible Microsampling Multi-omics for Precision Medicine. The iPOP Summit 2023**. 2023/4, Stanford University, USA.
10. **Bioinformatics Algorithm Development for Mass Spectrometry Data and its Application to Precision Medicine. University of Colorado**. 2023/4, Denver, Colorado, USA.
11. **Bioinformatics Algorithm Development for Mass Spectrometry Data and its Application to Precision Medicine. University of Minnesota**. 2023/4, Twin Cities, Minnesota, USA.
12. **Bioinformatics Algorithm Development for Mass Spectrometry Data and its Application to Precision Medicine. University of Connecticut**. 2023/3, Storrs, Connecticut, USA.
13. **Bioinformatics Algorithm Development for Mass Spectrometry Data and its Application to Precision Medicine. Altos Labs**. 2023/3, Redwood City, USA.
14. **Bioinformatics Algorithm Development for Mass Spectrometry Data and its Application to Precision Medicine. Ohio State University**. 2023/2, Columbus, USA.
15. **Bioinformatics Method Development for Mass Spectrometry and its Application to Precision Medicine. Stanford CVI Early Career Research Roundtable**, 2022/11, Stanford, USA.

Oral Presentations

1. **Molecular Phenotypes and Responses During Human Pregnancy.** ASMS 2025, 2025/6/5, Baltimore, USA.
2. **Nonlinear dynamics of multi-omics profiles during human aging.** Biomarkers of Aging Conference, 2024/10, Boston, USA.
3. **Nonlinear Dynamic Changes During Human Aging Revealed in Multi-omics Profiles.** Bay Area Metabolism Meeting (BAMM) 2023, 2023/9, Palo Alto, USA.
4. **Multi-omics Microsampling for The Profiling of Lifestyle-Associate Changes in Health.** The 3rd Chinese American Society for Mass Spectrometry Conference, 2023/8, Virtual meeting.
5. **Multi-omics Microsampling for The Profiling of Lifestyle-Associate Changes in Health.** The 71th American Society for Mass Spectrometry Conference, 2023/6, Houston, USA.
6. **Multi-omics Microsampling for The Profiling of Lifestyle-Associate Changes in Health.** Invited presentation, 2023/2, Huazhong University of Science and Technology, China.
7. **Multi-omics Microsampling for The Profiling of Lifestyle-Associate Changes in Health.** Stanford Postdoc Symposium, 2023/2, Stanford, USA.
8. **Deep Learning-based Pseudo-Mass Spectrometry Imaging Analysis for Precision Medicine.** CASMS Virtual Conference, 2022/10, Virtual meeting.
9. **TidyMass: An Object-oriented Reproducible Analysis Framework for LC-MS Data.** ACS Fall 2022, 2022/8, Virtual meeting.
10. **metID: an R Package for Automatable Compound Annotation for LC-MS-based Data.** The 69th American Society for Mass Spectrometry Conference, 2021/11, Philadelphia, USA.
11. **Decoding Links Between the Exposome and Health Outcomes by Multi-omics Analysis.** Exposome Data Challenge Event, 2021/4, Virtual meeting.
12. **Metabolic Reaction Network-based Recursive Metabolite Identification for Untargeted Metabolomics.** The 14th International Conference of the Metabolomics Society, 2018/6, Seattle, USA.
13. **Assessment of the Response to Neoadjuvant Chemo-Radiation in Rectal Cancer Patients based on a Metabolomics Approach.** The 3rd China Mass Spectrometry Analysis Conference, 2017/12, Xiamen, China.
14. **Normalization and Integration of Large-Scale Mass Spectrometry-based Metabolomics Data Using Support Vector Regression.** The 34th China Mass Spectrometry Society Conference, 2016/9, Xining, China.
15. **Normalization and Integration of Large-Scale MS-based Metabolomics Data Using Support Vector Regression.** The 64th American Society for Mass Spectrometry Conference, 2016/6, San Antonio, USA.

Poster Presentations

1. **Nonlinear Dynamic Changes During Human Aging Revealed in Multi-omics Profiles.** The 72nd American Society for Mass Spectrometry Conference, 2024/6, Anaheim, California, USA.
2. **Nonlinear Dynamic Changes During Human Aging Revealed in Multi-omics Profiles.** Stanford Bio-X Interdisciplinary Initiatives Seed Grants Program Symposium 2023. 2023/9, Stanford, USA.

3. **TidyMass An Object-oriented Reproducible Analysis Framework for LC-MS Data.** Stanford Data Science Conference, 2023/5, Stanford University, USA.
4. **Deep Learning-based Pseudo-Mass Spectrometry Imaging Analysis for Precision Medicine.** Stanford Genetics Retreat 2022, 2022/9, Monterey, USA.
5. **Deep Learning-based Pseudo-Mass Spectrometry Imaging Analysis for Precision Medicine.** Bay Area Metabolism Meeting (BAMM) 2022, 2022/9, Palo Alto, USA.
6. **TidyMass An Object-oriented Reproducible Analysis Framework for LC-MS Data.** The 70th American Society for Mass Spectrometry Conference, 2022/6, Minneapolis, USA.
7. **Longitudinal Interactions Between Levels of Serum Cytokine and the Microbiome from Four Body Sites.** IMMUNOLOGY2022, 2022/5, Portland, USA.
8. **TidyMass An Object-oriented Reproducible Analysis Framework for LC-MS Data.** Stanford Genetics Retreat 2021, 2021/9, Palo Alto, USA.
9. **Longitudinal Urine Metabolic Profiling and Gestational Age Prediction in Pregnancy.** The 17th International Conference of the Metabolomics Society, 2020/6, Virtual meeting.
10. **Metabolic Reaction Network based Metabolite Annotation in Untargeted Metabolomics.** The 13th International Conference of the Metabolomics Society, 2017/6, Brisbane, Austria.

Teaching

- **Bioanalytical Techniques** 2025/10-Present, **Nanyang Technological University, Singapore**
- **Introduction to Biostatistics with R** 2025/3-Present, **Nanyang Technological University, Singapore**
- **LC-MS Data Processing and Analysis Using R language** 2022/9, **Biotree Company**
Responsibilities: Lecture
- **R Language for Reproducible Data Analysis** 2020/4, **SCPA**
Responsibilities: Lecture
- **Stanford School of Medicine Intern** 2019/8–2024/5, **Stanford University**
Responsibilities: Mentor
- **Mass Spectrometry Analysis** 2014/9–2014/12, **Chinese Academy of Sciences**
Responsibilities: Guest lecturer

Volunteering

- **Member of Editorial Board** 2023/8-present, **[npj Women's health](#)**
- **Member of Youth Editorial Board** 2023/2–present, **[Phenomics](#)**
- **Member of Youth Editorial Board** 2023/1–present, **[Brain-X](#)**
- **Member of Youth Editorial Board** 2022/7–present, **[iMeta](#)**
- **Co-chair** 2021/7–2022/8, **[Stanford Chinese Postdoctoral Association](#)**
- **Organizer** 2020/2–2021/12, **[Stanford Chinese Postdoctoral Oral Presentation Improvement](#)**
- **Member** 2019/7–2021/7, **[Stanford Chinese Postdoctoral Association](#)**

Journal Referee

- *Nature Biotechnology, Nature Aging, Nature Methods, Nature Genetics, Nature Communications, Cell Metabolism, Briefing in Bioinformatics, Bioinformatics, GigaScience, PLOS one, Genomics, Proteomics & Bioinformatics, Communications Chemistry, Communications Biology, Frontiers Molecular Biosciences, iMeta, iMetaOmics, Proteomics, BMC Bioinformatics, Proteomics, Cell Systems, Journal of Integrative Plant Biology, Cell Reports Medicine, Cell Reports Methods, Advanced Science*

Referrers

- **Professor Michael Snyder**
Postdoctoral Advisor
Stanford University, USA
Phone number: +1 (650) 723-4668
Email: mpsnyder@stanford.edu
- **Professor Zheng-Jiang Zhu**
Ph.D. Advisor
Director of Metabolomics Research Center, IRCBC, **Chinese Academy of Sciences, China**
Phone number: +86 21-68582296
Email: jiangzhu@sioc.ac.cn
- **Professor Caroline Johnson**
Yale University, USA
Phone number: +1 (203) 785-2882
Email: caroline.johnson@yale.edu
- **Professor Mingliang Fang**
Fudan University, China
Email: mlfang@fudan.edu.cn
- **Professor Anne Brunet**
Stanford University, USA
Email: abrunet1@stanford.edu
- **Professor Pieter C. Dorrestein**
University of California San Diego, USA
Email: pdorrestein@health.ucsd.edu
- **Professor Shahar Lev-Ari**
Tel-Aviv University, Israel
Email: slevari@stanford.edu